

Report: an AI support agent for AmazonHelp that knows when to shut up

1. Problem framing: what "good" means

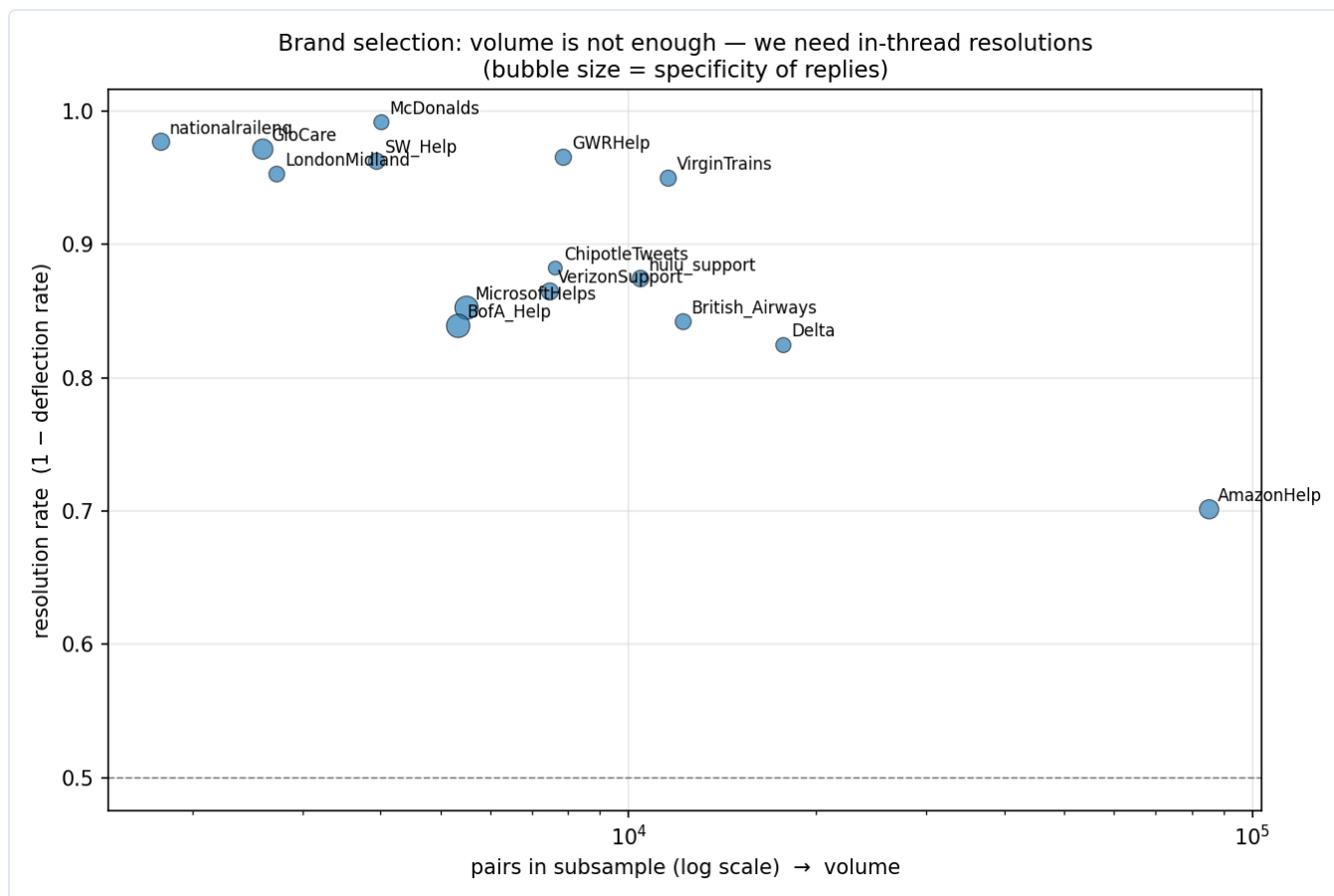
Good = **safe automation, not maximum automation**. For AmazonHelp, a wrong auto-handled reply (invented refund date, wrong troubleshooting, missed fraud) costs far more than an unnecessary escalation costs an agent's minute. So the headline result is not accuracy but the **coverage-vs-unsafe curve**: how much routine work we auto-handle at each level of false auto-handling of must-escalate cases.

The agent does three things per message: (1) classify into 5 operational intents + `other_unclear`, (2) draft a reply grounded ONLY in retrieved past AmazonHelp resolutions, (3) emit auto/human + a machine-readable reason code.

Non-goals: multi-turn dialogue memory; fine-tuning; multilingual handling (English-only by explicit filter); all-brand coverage (one brand, deep); live posting (draft/approve only); Qdrant Cloud in the eval path.

2. Data & brand selection

Primary: `twcs.csv` (~3M tweets). Rebuilt (customer → brand reply) pairs with follow-ups: 510,927 pairs across 108 brands from a 1.2M-row scan. Working set: 6,000 AmazonHelp pairs. Chose **AmazonHelp** (85k pairs, largest absolute grounding pool) with eyes open: rank 14 on resolution *rate* (70.1%, 29.9% deflections, 19% non-English) but #1 on groundable volume (48.6k). Table: `data/sample/brand_profiles.csv`; chart:



;
method: docs/brand_selection.md .

Key trap found and measured: most big brands "resolve" by replying "please DM us" — a grounding corpus of deflections teaches the generator to deflect while scoring perfectly on groundedness. We filter deflections (URL-aware, self-service override, data-mined patterns) OUT of the retrieval corpus, and score deflection ONLY over English pairs (Spanish "DM us" matches no English regex; without this AmazonHelp scored a bogus 96.9% resolution rate).

Gratitude check failed honestly: customers thank deflections slightly MORE (+compliant "thanks, will DM"), lift ≈ 0 even after cleaning. Gratitude is politeness, not ground truth — kept as diagnostic only.

Leakage control (§6 of the build): golden ids AND exact texts are excluded from both the retrieval corpus (3,334 cases) and the TF-IDF weak-train pool; scripts/check_leakage.py fails loudly (exact 0, near-dupe 0, train 0 — verified). The audit caught the pre-fix state leaking 100/150 golden messages into the corpus and 108/150 into B1's training pool; B1's macro-F1 fell 0.644 $\rightarrow$ 0.354 after cleaning. That collapse is reported, not hidden.

3. System

message $\rightarrow$ intent (Gemini 3.1-flash-lite, few-shot, JSON) $\rightarrow$ Qdrant embedded retrieval (top-3 of 3,334 English non-deflection cases) $\rightarrow$ risk rules $\rightarrow$ RAG draft $\rightarrow$ code grounding validator (cited ids must exist; promise patterns must be verbatim in evidence) $\rightarrow$ gate . Gate autos ONLY on: known intent + score + similarity + **intent-evidence agreement $\geq 2/3$** + low risk + validator pass. model_score is an uncalibrated score, never a probability (measured: wrong intents average 0.93 — the confidence threshold is empirically dead weight;

agreement does the real filtering). Retrieval intent-consistency: 0.47 / 0.63 / 0.74 at K=1/3/5; self-hits 0.

4. Results vs baselines (frozen test-100; thresholds from frozen cal-50)

system	intent acc [95% CI]	macro-F1 [95% CI]	esc P / R	unsafe	coverage
B0 trivial (measured-majority <code>other_unclear</code> + canned + always-escalate)	0.090 [0.04, 0.15]	0.028 [0.01, 0.04]	0.23 / 1.00	0.000	0.00
B1 simple (TF-IDF LogReg + verbatim retrieval, golden-excluded train)	0.340 [0.25, 0.43]	0.354 [0.25, 0.44]	0.23 / 0.96	0.043	0.04
Ours (LLM + RAG + agreement gate)	0.720 [0.63, 0.80]	0.676 [0.56, 0.77]	0.28 / 1.00	0.000	0.17

Weighted F1: 0.015 / 0.371 / 0.711. Confusion matrices saved in `outputs/eval_results.json` (ours: only miss pattern of note is delay↔missing, 9 cases). Escalation confusion (ours): 23 caught / 60 over-escalated / 0 missed / 17 correctly auto-handled. Reason codes: WEAK_EVIDENCE 51, HIGH_RISK 18, OK_AUTO 17, UNKNOWN_INTENT 13, GROUNDING_FAIL 1 — the gate over-escalates by design (precision 0.28 is the price of recall 1.00).

Per-class (ours): refund F1 0.83 (n=27), device 0.84 (n=11), delay 0.75 (n=27), other 0.64 (n=9), missing 0.53 (n=19), status 0.46 (n=7). Status/other CIs span ±0.2 — unmeasurable at this n, stated not hidden.

Central result — coverage-vs-unsafe on cal-50 (full table `outputs/coverage_curve.json`, regenerated by every `make eval`): (0.34 cov, 0.231 unsafe) → operating point (0.22, 0.154) → test at the point: (0.17, 0.000). The table, not the 0.72 accuracy, is the claim — grid rows collapse to 2 distinct operating regimes because the confidence knob is empirically dead (scores are bimodal ~0/0.9), which is itself a finding. Gate ablation (no new calls, contract-§18 definition throughout): dropping agreement → coverage 0.48 at unsafe 0.174 (4 missed); dropping sim too → 0.65/0.304 (7 missed). Agreement is what makes automation safe. A 0-miss test (0 of 23) has an upper CI of ~0.13 (rule of three) — zero is observed, not proven.

Reply quality (human-primary by design): ours 4.16 (n=50) vs B1 verbatim 2.45 (n=20) vs B0 canned 2.70 (n=10), same 1–5 groundedness rubric. Verbatim copies score low because real brand replies are often fragments ("(3/3)" tails) or wrong-context; canned is safe but generic. Deterministic checks on all 100: grounding pass 0.95–0.99, evidence cited 0.99, promise-pattern violations 1 (a legit "\$1 authorization hold" the regex over-fires on — §5 mode 3). LLM judge (gemini-3.1-flash-lite, blinded, structured JSON): banked live on 98 ours-drafts (judge_avg 4.314) and validated against the human on the 50 paired drafts: Spearman ρ =0.173, binned κ =-0.11, raw agreement 0.36, mean|diff|=0.675 (judge mean 4.20 vs human 4.16). Means agree; per-item ranking does not. Two stated causes: restriction of range (RAG drafts cluster 3–5) and judge generosity on borderline drafts. Consequence: judge is a secondary cross-check only — humans stay primary; baseline drafts un-judged (quota), stated not hidden (`outputs/judge_agreement.json` has all pairs).

Cost/latency: 436 cached calls, ~244k tokens, \$0 actual spend (free tier; ~\$0.07 at paid list). Retrieval p50 32ms / p95 39ms local; LLM classify ~2–4s, drafts similar (observed ranges).

5. Failure analysis (top 5, 28 intent failures, all real)

`results/failure_analysis.md` has each with frequency, expected vs actual, real example, why, hypothesis, fix. Summary:

1. **Delay↔missing boundary (9/28, 32%)** — late vs lost share vocabulary (amz-012, 082, 005...); rules, LLM, and annotators all split here. Fix: "failed attempt" third state.
2. **Cancelled orders over-trigger refund (4/28)** — amz-067/111/023/134: model over-applies the refund-priority rule where gold wants status. Fix: refund only on explicit money words.
3. **Noisy rants punted to other (5/28)** — amz-128/031/096/098/088: safe direction (all escalated) but intent recall suffers; few-shots lack rants.
4. **Tracking/returns misrouted (4/28)** — amz-119/028/125/009: carrier and return-to-sender vocabulary sits equidistant from three intents.
5. **Residual (7/28)** — genuinely ambiguous amz-142 ("when is my echo coming?", device vs delivery); validator regex over-fire on amz-051 "\$1 authorization hold". The prior escalation miss amz-102 ("No tracking???", 12 chars) is FIXED: the ultra-short floor now escalates it (3 messages caught by the floor on test-100).

6. What is misleading about my headline number? (mandatory)

- Macro-F1 0.676: delay+refund are 54% of test; status (n=7)/other (n=9) F1s are noise (CI table above). The per-class row, not the headline, is honest.
- Unsafe 0.000 = 0 missed of 23 must-escalate; with n=23 the 95% upper CI is ~0.13 (rule of three). Zero misses is observed, not proven. The curve is the claim.
- Flash-lite outputs vary run-to-run even at temperature 0: a live re-run measured cal coverage 0.40 vs replay-pinned 0.22. Replay cache pins exact numbers; live numbers wobble. Reported numbers are the pinned ones.
- Golden: 4 adjudication passes by ONE annotator; delay↔missing labels moved twice. Some "model errors" are annotator noise (30 rows flagged for a second annotator; pending). Test set is frozen now — earlier numbers in git history refer to shifted splits and are superseded.
- The B1 collapse (0.644 → 0.354) cuts both ways: it proves the leakage audit mattered, but it also means our margin over "simple" partly reflects B1's training-data discipline, not just modeling.
- Judge shares the generator's vendor AND validates weakly against the human: paired n=50 gives $p=0.173$, $\kappa=-0.11$. Judge_avg 4.314 is a mean, not a ranking instrument — human scores carry every reply-quality claim.
- One brand, Oct-2017 window, English-only, n=100: generalises nowhere untouched. Coverage 0.17 means 5 of 6 messages still need a human.

7. One more week

1. Second annotator: 30 flagged double-labels → noise-bounded metrics and a real inter-rater κ ; extend judge agreement to baseline drafts + probe a stronger judge model (agreement is the weak link, not the drafts).
2. "Failed attempt" intent (delay↔missing is 31% of failures); port payment-complexity and batch-question patterns into the gate.
3. Promise-regex → NLI-vs-cited-span validator (kills the \$1 over-fire).
4. Bigger cal split (100) + frozen re-run; Lite run-to-run variance study.
5. Shadow-mode approve/edit UI → active-learning loop from human edits.

8. Decision log

`DECISIONS.md` holds exactly 15 entries (WHAT/WHY/alternatives/tradeoff) — the ones above, compressed. Overflow detail lives in `GOLDEN_NOTE.md`.